Exercise 7

On cubic roots of unity

From the school book | Understand | Apply | Higher Order Thinking Skills

1 Choose the correct answer from those given :

(1) $\omega^{22} + \omega^{32} =$ .....

(a) ω (b) 1 (c) -1 (d) zero

(2) $\omega^{-13} + \frac{1}{\omega} =$ .....

(a) ω^2 (b) 1 (c) $2\,\omega^2$ (d) $2\,\omega$

(3) $\omega^{2020} + \omega^{2021} + \omega^{2022} =$ .....

(a) ω (b) ω^2 (c) zero (d) 1

(4) $(\omega - \omega^2)^4 =$ .....

(a) $\pm\sqrt{3}\,i$ (b) $3\,i$ (c) 3 (d) 9

(5) $\left(\omega + \frac{1}{\omega}\right)^2 \left(\omega^2 + \frac{1}{\omega^2}\right)^2 =$ .....

(a) zero (b) 1 (c) -1 (d) ω^2

(6) $(1 + \omega)^4 + (1 + \omega^2)^4 + (\omega + \omega^2)^4 =$ .....

(a) zero (b) 1 (c) -1 (d) ω

(7) $\left(\frac{1}{\omega + 1}\right)\left(1 + \omega - \frac{3}{\omega}\right) =$ .....

(a) zero (b) 1 (c) 2 (d) 4

(8) If $\mathcal{X} = \frac{-1 + \sqrt{3}\,i}{2}$ where $i^2 = -1$, then $\mathcal{X}^5 + \mathcal{X}^4 + 5 =$ .....

(a) -1 (b) 1 (c) ω (d) 4

(9) The two numbers that each is the square of the other are .....

(a) $1 , -1$ (b) $i , -i$ (c) $\omega , -\omega$ (d) ω , ω^2

(10) If a , b , c are three consecutive integers , then $\omega^a + \omega^b + \omega^c =$ .....

(a) zero (b) 1 (c) ω (d) ω^2

(11) The conjugate of the number ω equals .....

(a) ω (b) ω^2 (c) 1 (d) $-\omega$

(12) The conjugate of the number $1 + \omega$ is .....

(a) $1 - \omega$ (b) $-\omega^2$ (c) $1 + \omega^2$ (d) $1 - \omega^2$

(13) The conjugate of the number $i - \omega^2$ is .....

(a) $i + \omega^2$ (b) $-i + \omega^2$ (c) $-i - \omega$ (d) $-i + \omega$

(14) The conjugate of the complex number $(i + \omega) + 2023$ is the complex number .....

(a) $-i + \omega + 2023$ (b) $-i + \omega^2 + 2023$

(c) $-i - \omega - 2023$ (d) $-i - \omega^2 - 2023$

(15) The conjugate of the number $2\omega + 3\omega^2$ is .....

(a) $2\omega - 3\omega^2$ (b) $2\omega^2 + 3\omega$ (c) $3\omega - 2\omega^2$ (d) $3\omega + 3\omega^2$

(16) $\frac{3}{2+\omega} =$ .....

(a) $2 - \omega$ (b) $2 + \omega^2$ (c) $3(1 + \omega)^2$ (d) $3\omega^2$

(17) $\frac{a + b\omega + c\omega^2}{c + a\omega + b\omega^2} + \frac{a + b\omega + c\omega^2}{b + c\omega + a\omega^2} =$ .....

(a) -1 (b) zero (c) 1 (d) 2

(18) The solution set of $x^3 = 8$ in $\mathbb{C}$ is .....

(a) $\{2\}$ (b) $\{2, 2\omega, 4\omega^2\}$

(c) $\{2, 2\omega, 2\omega^2\}$ (d) $\{8, 8 + \omega, 8 + \omega^2\}$

(19) *(2nd Session 2017)* $1 + \omega + \omega^2 + \omega^3 + \cdots + \omega^{100} =$ .....

(a) zero (b) 1 (c) ω (d) $-\omega^2$

(20) $\left(\omega^2 + \frac{1}{\omega}\right)\left(1 + \frac{1}{\omega^2}\right)^2 =$ .....

(a) 2 (b) zero (c) -3 (d) -5

(21) Which of the following results belongs to the real numbers ?

(a) $\frac{1 + \omega}{1 + \omega^2}$ (b) $(1 + \omega)(2 + \omega^2)$

(c) $(3 + \omega)(3 + \omega^2)$ (d) $(\omega^2 - \omega)^3$

(22) If z_1 , z_2 are conjugate numbers and $z_1 = (c + d)\,\omega^2 + (c - d)\,\omega$, $z_2 = 7\,\omega^2 + 5\,\omega$, then $c \times d =$ .....

(a) 7 (b) -5 (c) -4 (d) -6

(23) If $\mathcal{X} = a$, $y = b\,\omega$, $z = c\,\omega^2$, then $\frac{\mathcal{X}}{a} + \frac{y^2}{b^2} + \frac{z^2}{c^2} =$ .....

(a) zero (b) 1 (c) 2 (d) ω

(24) $\cos\left[(\omega^{10} + \omega^{23})\,\pi + \frac{\pi}{4}\right] =$ .....

(a) $\frac{-\sqrt{3}}{2}$ (b) $\frac{-\sqrt{2}}{2}$ (c) $\frac{\sqrt{2}}{2}$ (d) $\frac{\sqrt{3}}{2}$

(25) If $z = 2 + 3\,\omega^2$, then $z\,\bar{z} =$ .....

(a) 4 (b) 6 (c) 7 (d) 9

(26) *(Trial 2021)* $\left(\frac{a}{\omega} - \frac{a}{\omega^2} + \frac{3a}{\omega^4} - \frac{3a}{\omega^5}\right)^2 =$ .....

(a) $-48\,a^2$ (b) $48\,a^2$ (c) $16\,a^2$ (d) $-16\,a^2$

(27) *(1st Session 2021)* If 1 , ω , ω^2 are the cubic roots of unity , and $\mathcal{X} = \frac{1}{1 + \omega\,i}$, $y = \frac{\omega + i}{1 + \omega^2\,i}$, then $\mathcal{X} - y =$ .....

(a) $i + 1$ (b) $1 - i$ (c) 1 (d) i

(28) *(1st Session 2023)* If 1 , ω and ω^2 are the cubic roots of unity , then $\left(\frac{a}{\omega} - a + a\,\omega^7\right)^4 =$ ..... , (where $a \neq 0$)

(a) $8\,a^4$ (b) $16\,a^4$ (c) $6\,a^4$ (d) $2\,a^4$

(29) *(2nd Session 2023)* If z is a complex number , and ω is one of the cubic roots of unity , and $z\,(1 - \omega - \omega^2) = \frac{2}{1 + \omega}$, then $z =$ .....

(a) $-\omega$ (b) $1 + \omega$ (c) $1 - \omega^2$ (d) ω

2 If 1 , ω , ω^2 are the cubic roots of unity , find the value of each of the following :

(1) $(1 - \omega)\,(1 - \omega^2)$

(2) $5 + 3\,\omega + 3\,\omega^2$

(3) $5 + 5\,\omega - 5\,\omega^2$

(4) $(2 + 5\,\omega + 2\,\omega^2)^5$

(5) $(1 - 2\,\omega + \omega^2)\,(1 + \omega - \omega^2)$

(6) $(1 + \omega^2)^{15}$

(7) $(1 + \omega + 2\,\omega^2)^2 + (1 + 2\,\omega + \omega^2)^2$

(8) $(1 + \omega)^3 + (1 - \omega - \omega^2)^3$

(9) $(1 - \omega)\,(1 - \omega^2)\,(1 + \omega^3)$

(10) $\left(1 - \frac{2}{\omega} - \frac{2}{\omega^2}\right)(3 - 5\,\omega - 5\,\omega^2)$

(11) $\left(\omega + \frac{1}{\omega}\right)^2 \left(\omega^2 + \frac{1}{\omega^2}\right)^2$

(12) $\left(1 + \frac{1}{\omega} + i\right)\left(1 + \frac{1}{\omega^2} + i\right)$

3 *(1st Session 2005)* The 1 , ω and ω^2 are the cubic roots of unity and a , b , c , d are real numbers , find the numerical value of the expression : $\frac{a + b\,\omega}{a\,\omega^2 + b} + \frac{c\,\omega^2 + d}{c + d\,\omega}$ « – 1 »

4 **If 1 , ω , ω^2 are the cubic roots of unity , prove each of the following identities :**

(1) $(1 + \omega^2)^8 = \omega^2$

(2) $\left(1 - \frac{2}{\omega^2} + \omega^2\right)\left(1 + \omega - \frac{5}{\omega}\right) = 18$

(3) $(1 - \omega^2 + \omega^4)^2 + (1 + \omega^2 + \omega^4)^2 = 4\,\omega$

(4) $\frac{5 - 4\,\omega + 11\,\omega^2}{1 - 2\,\omega + 3\,\omega^2} = 3$

(5) *(1st Session 2011)* $\left(\frac{1}{\omega^2\,(\omega + 3)} - \frac{1}{1 + 3\,\omega^4}\right)^2 = \frac{-27}{49}$

(6) $\frac{1}{4 + 3\,\omega + 2\,\omega^2} + \frac{1}{5 + 3\,\omega + 4\,\omega^2} = 1$

(7) $\left(\frac{\omega}{1 + 2\,\omega}\right)^2 + \left(\frac{1 + 2\,\omega^2}{\omega^2}\right)^2 = \frac{-10}{3}\,\omega^2$

(8) $\left[\frac{1}{1 + \omega\,i} - \frac{\omega + i}{1 + \omega^2\,i}\right]^8 = 16$

(9) $\frac{\omega^2\,(\omega - 1)\,(\omega^2 - 1)}{(2\,\omega + 1)\,(\omega^2 + 2)} = -1$

(10) *(1st Session 2014)* $\left[\frac{5 - 3\,\omega^2}{5\,\omega - 3} - \frac{2 - 7\,\omega}{2\,\omega^2 - 7}\right]^4 = 9$

(11) $\frac{3 + 5\,\omega + 3\,\omega^2}{1 - 2\,\omega - 4\,\omega^2} + \frac{3 + 5\,\omega^2 + 3\,\omega}{1 - 2\,\omega^2 - 4\,\omega} = \frac{-2}{19}$

(12) $\left(\frac{4 + \omega^2 + 2\,\omega}{\omega^2 - 2} + \frac{\omega - 1}{3 + \omega + 2\,\omega^2}\right)^2 = 4$

(13) $\frac{5 + 3\,\omega^2}{2 - \omega} + \frac{5 + 3\,\omega}{2 - \omega^2} = \frac{22}{7}$

(14) $(5 + 4\,\omega)^3 \times (5 + 4\,\omega^2)^3 = 9261$

(15) $\left(\frac{\omega}{2 + 3\,\omega} - \frac{\omega^2}{2 + 3\,\omega^2}\right)^2 = \frac{-12}{49}$

(16) $\left(\frac{1}{1 + 2\,\omega^2} - \frac{1}{1 + 2\,\omega}\right)^4 = \frac{16}{9}$

(17) $\left(5 - \frac{5}{1 + \omega^2} + \frac{3}{\omega^2}\right)^6 = 64$

(18) $\left[\frac{a + \omega\,b + \omega^2\,c}{\omega^2\,a + b + c\,\omega} - \frac{c + a\,\omega^2 + b\,\omega}{\omega\,c + b\,\omega^2 + a}\right]^8 = 81$

(19) $\left(x+\frac{y}{\omega}\right)\left(x+\frac{y}{\omega^2}\right)=(x-y)^2+xy$

(20) *(1st Session 2015)* $(a+b\,\omega+a\,\omega^2)(a+b\,\omega^2+a\,\omega^4)=(b-a)^2$

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5 **If $1, \omega, \omega^2$ are the cubic roots of one, prove that :**

(1) $(1+\omega)(1+\omega^2)(1+\omega^3)(1+\omega^4)(1+\omega^5)=2$

(2) $(1-\omega)(1-\omega^2)(1-\omega^4)(1-\omega^5)=9$

(3) $(2+3\,\omega^2)\left(2+\frac{3}{\omega^2}\right)\left(3+\frac{4}{\omega}\right)(3+4\,\omega)=91$

(4) 📖 $(1-\omega+\omega^2)(1-\omega^2+\omega^4)(1-\omega^4+\omega^8)(1-\omega^8+\omega^{16})=2^4$

(5) $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)$ ... to 2 n factors $=1$

(6) $(1-\omega)(1-\omega^2)(1-\omega^4)(1-\omega^8)$ ... to 2 n factors $=3^n$

(7) $(2+3\,\omega+2\,\omega^2)(2+3\,\omega^2+2\,\omega^4)(2+3\,\omega^4+2\,\omega^8)$ ... to 2 n factors $=1$

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6 *(2nd Session 2019)* Find in the simplest form the value of the expression :

$\left[k-\frac{k-1}{1+\omega}+(k+1)\omega^2\right]^8$, where $k\in\mathbb{R}$

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7 *(2nd Session 2002)* If $\left(\frac{\omega^2}{1+2\,\omega^2}\right)^4+\left(\frac{1+2\,\omega}{\omega}\right)^4=a\,\omega^2$, find the value of a «$\frac{82}{9}$»

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8 If $1, \omega, \omega^2$ are the cubic roots of unity :

Prove that : $\left(\frac{a+1}{a+(2a+1)\,\omega+a\,\omega^2}\right)^6$ does not depend on the value of a

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9 *(2nd Session 2018)* If $\frac{1+10\,\omega+10\,\omega^2}{1-3\,\omega-3\,\omega^2}=(k\,i)^2$, find the value of the real number k «$\pm\frac{3}{2}$»

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10 If $x=2\,\omega-3\,\omega^2$, $y=3+5\,\omega^2$

find the value of each : $x+y$, xy , **then prove that :** $\frac{x^3+y^3}{x^2+y^2}=\frac{56}{37}$ «1 , 19»

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11 If $\omega=\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}$, **prove that :** $(1+\omega)^n=\cos\frac{n\pi}{3}+i\sin\frac{n\pi}{3}$

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12 📖 **Prove that :** $x=\frac{-1+\sqrt{-3}}{2}$ is one of the roots of the equation : $x^{10}+x^5+1=0$

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13 📖 If $x=\frac{-1+\sqrt{3}\,i}{2}$, **prove that :** $x^6+6x^5+15x^4+20x^3+15x^2+6x=0$

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14 **Prove that :** ω^2 is one of the roots of the equation : $3x^{11}+2x^8+5x^7+5=0$

15 Choose the correct answer from those given :

(1) $\left(2+\frac{3}{\omega}\right)\left(2+\frac{3}{\omega^2}\right)\left(3-\frac{2}{\omega}\right)\left(3-\frac{2}{\omega^2}\right)$ = ...

(a) 7 (b) 9 (c) 26 (d) 133

(2) $\left(1-\frac{1}{\omega}\right)\left(1-\frac{1}{\omega^2}\right)\left(1-\frac{1}{\omega^4}\right)\left(1-\frac{1}{\omega^8}\right)$ ... to 10 factors = ...

(a) 243 (b) 15 (c) 5 (d) – 5

(3) $\left(\frac{3+5\omega}{5+3\omega^2}-\frac{5+3\omega^2}{3+5\omega}\right)^8$ = ...

(a) 1 (b) 27 (c) 81 (d) 243

(4) If $a = 2\omega - 3\omega^2$, $b = 3 + 5\omega^2$, then $a^2 + b^2$ = ...

(a) – 37 (b) – 19 (c) 1 (d) 38

(5) $(a + b\omega + a\omega^2)(a + b\omega^2 + a\omega^4)$ = ...

(a) 1 (b) a – b (c) $(a - b)^2$ (d) $b^2 - a^2$

(6) *(2nd Session 2018)* $\left(1 + 2\omega^5 + \frac{1}{\omega^2}\right)\left(1 + 2\omega + \frac{1}{\omega^4}\right)$ = ...

(a) 0 (b) 1 (c) –1 (d) 2

(7) *(2nd Session 2019)* $\frac{a - d\omega}{a\omega^2 - d} - \omega^2$ = ...

(a) 3 i (b) $\pm\sqrt{3}\,i$ (c) –3 (d) 3

(8) *(1st Session 2019)* If $z = \omega^{\mathcal{X}}$, where $\mathcal{X}$ is a positive integer , then $|z|$ = ...

(a) 1 (b) ω (c) $\mathcal{X}$ (d) ω^2

(9) $\sum_{r=1}^{5} \omega^r$ = ...

(a) zero (b) – 1 (c) 1 (d) ω^2

(10) $\sum_{r=1}^{6} (1 + \omega^r)$ = ...

(a) zero (b) 6 (c) 1 (d) $1 + \omega$

(11) $1 + \sum_{r=1}^{6} \omega^r$ = ...

(a) zero (b) 1 (c) ω (d) $2 - \omega^2$

(12) If $i^{12} = \omega^{12}$, then .....

(a) $i = \omega$

(b) $i = \pm\omega$

(c) i , ω each is a root of the equation $z^{12} = 1$

(d) i , ω have no relation.

(13) *(2nd Session 2018)* If $(1 + \omega)^7 = a + b\,\omega$ where a and b are real numbers , then $(a , b) =$ .....

(a) $(0 , -1)$ (b) $(1 , 1)$ (c) $(0 , 1)$ (d) $(1 , -1)$

(14) If $(1 + \omega^2)^n = (1 + \omega)^n$, then the least value of the positive integer n is .....

(a) 2 (b) 3 (c) 5 (d) 6

(15) If a , b are the non real cubic roots of unity , then $a\,b + a^5 + b^5 =$ .....

(a) 1 (b) zero (c) -1 (d) 3

(16) If 3 , $2 + \omega^2$ are two roots of cubic equation that has real coefficient , then the third root equals .....

(a) $2 + \omega$ (b) $2 - \omega$ (c) $\omega^2 - 2$ (d) $2 - \omega^2$

(17) The value of the number $(-1 + \sqrt{3}\,i)^{48}$ where $i^2 = -1$ equals .....

(a) 1 (b) 2 (c) 2^{24} (d) 2^{48}

(18) $\left(\frac{-1 - \sqrt{3}\,i}{2}\right)^{3n} + \left(\frac{-1 + \sqrt{3}\,i}{2}\right)^{3n} =$ ..... where $n \in \mathbb{Z}$

(a) $\frac{3}{2}$ (b) 3 (c) zero (d) 2

(19) $\left(\frac{1}{2} + \frac{3}{8} + \frac{9}{32} + \frac{27}{128} + \ldots\right) + \omega + \omega^2 =$ .....

(a) -1 (b) 1 (c) $-i$ (d) i

(20) The product of the cubic roots of the real number $\mathcal{X}$ equals .....

(a) $\mathcal{X}$ (b) $\omega\,\mathcal{X}$ (c) $\omega^2\,\mathcal{X}^2$ (d) $\mathcal{X}^3$

(21) $\left(\frac{-1 - \sqrt{-3}}{2}\right)^{22} + \left(\frac{-1 - \sqrt{-3}}{2}\right)^{17} =$ .....

(a) zero (b) -1 (c) 1 (d) ω

(22) If z_1 , z_2 , z_3 are complex numbers represented by vertices of an equilateral triangle whose centroid is the origin , then $z_1 + z_2 + z_3 =$ .....

(a) zero (b) 1 (c) ω (d) ω^2

(23) If $\mathcal{X} + \frac{1}{\mathcal{X}} = -1$, then $\mathcal{X}^{2018} + \mathcal{X}^{-2018} =$ .....

(a) -1 (b) zero (c) 1 (d) 2

(24) Sum of roots of the equation : $(z - 2)^3 = 1$ equals .....

(a) zero (b) 2 (c) 1 (d) 6

(25) If $\ell = a\,\omega + b\,\omega^2$, $m = a\,\omega^2 + b\,\omega$ are complex numbers where $a , b \in \mathbb{R}$, then which of the following statements is false ?

(a) ℓ , m each is a multiplicative inverse of the other.

(b) ℓ , m are conjugate numbers

(c) $\ell^2 - m^2 = \pm\sqrt{3}\,i\,(a^2 - b^2)$ (d) $\omega\,\ell + \omega^2\,m \in \mathbb{R}$

(26) *(Trial 2021)* $\dfrac{14 + 6\,\omega + 21\,\omega^2}{8\,\omega^2 - 7} =$ .....

(a) $-\omega$ (b) ω (c) ω^2 (d) $-\omega^2$

(27) $\frac{1}{2}$, $\frac{1}{2}\,\omega$, $\frac{1}{2}\,\omega^2$ are the cubic roots of the number .....

(a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{2}\,i$ (d) $\frac{1}{8}$

(28) 2 , $\frac{1}{2} + \frac{\sqrt{3}}{2}\,i$, $\frac{1}{2} - \frac{\sqrt{3}}{2}\,i$ are the cubic roots of the equation .....

(a) $z^3 = 1$ (b) $z^3 - 8 = 0$ (c) $(z - 1)^3 = 1$ (d) $(z + 1)^3 = 1$

(29) *(Trial 2021)* On Argand diagram , area of the circle passes through the points represent the cubic roots of unity equals ..... square units.

(a) π (b) $2\sqrt{3}\,\pi$ (c) $\sqrt{3}\,\pi$ (d) $2\,\pi$

(30) *(Trial 2021)* On Argand diagram , area of the triangle whose vertices are the points represent the cubic roots of one equals ..... square units.

(a) $\frac{3\sqrt{3}}{4}$ (b) $\frac{3\sqrt{3}}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{3}}{4}$

16 Find the two square roots of each of the two following numbers :

(1) ω **(2)** $1 + \omega$

17 Form the quadratic equation whose two roots are :

(1) $1 + \omega\,i$, $1 + \omega^2\,i$

(2) 📖 $\frac{1}{1+\omega}$, $\frac{1}{1+\omega^2}$

(3) 📖 *(2nd Session 2014)* $(1 + \omega - \omega^2)^3$, $(1 - \omega + \omega^2)^3$

(4) $(2\,\omega + 2\,\omega^2 - \omega^3)^3$, $(2 - 2\,\omega - 2\,\omega^2)^3$

(5) $\frac{3+\omega}{2-\omega}$, $\frac{3+\omega^2}{2-\omega^2}$

(6) $\left(1 - (1 + \omega)^{-1}\right)^{-1}$, $\left(1 - (1 + \omega^2)^{-1}\right)^{-1}$

18 ***(Speciman 2025)*** Find the quadratic equation whose coefficients are real numbers and one of its two roots is $(1 + \omega + 3\omega^2)^2$

19 ***(El Sudan 2015)*** Put the number $z = 2\omega^4 + 2\omega^5 + 3 + i\left(5\sqrt{3}\,\omega^3 + 4\sqrt{3}\,\omega^2 + 4\sqrt{3}\,\omega\right)$ in the form $x + y\,i$, then find the two square roots of the number z in the exponential form. « $\sqrt{2}\,e^{\frac{\pi}{6}i}$, $\sqrt{2}\,e^{\frac{-5\pi}{6}i}$ »

20 If $z = 2(\omega + i)(\omega^2 + i)$, find all different forms of the number z and the square roots of z in the trigonometric form.

21 Put in the exponential form the complex number $z = \frac{1+\omega}{(1-i)^2} + \frac{1-\omega}{(1+i)^2}$, where $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, then find the two square roots of the number z in the trigonometric form.

22 **Find the values of x , y where x , y are two real numbers in each of the following :**

(1) If $(1 - i)(x + y\,i) = 9\left(\frac{1}{2 - 3\omega^2} + \frac{1}{5 + 3\omega^2}\right)$ « $\frac{63}{38}$, $\frac{63}{38}$ »

(2) ***(1st Session 2018)*** If $(x + y\,i)(1 - 3\,i) = 37\left(\frac{1}{3 - 4\omega^2} + \frac{1}{7 + 4\omega^2}\right)$ « 1 , 3 »

(3) If $(x + y\,i)(2 + i)^2 = \frac{1}{(1+\omega)^2} + \frac{1}{(1+\omega^2)^2}$ « $\frac{-3}{25}$, $\frac{4}{25}$ »

23 **Solve each of the following equation where $x \in \mathbb{C}$:**

(1) $x^2 = 1 + \omega^2$

(2) $4x^2 - 4\omega x - \omega = 0$

(3) $(2x + 1)^6 - 2(2x + 1)^3 + 1 = 0$

(4) ***(2nd Session 2013)*** $(3x - 2)^6 - 9(3x - 2)^3 + 8 = 0$

24 **Prove that :** The two roots of the equation $x^2 - (2 - i)x - i = 0$ are $(1 + \omega\,i)$, $(1 + \omega^2\,i)$

25 **Prove that :** $(1 - \omega^2)^{3n} = (\omega - 1)^{3n}$ where : n is a positive odd integer.

26 Find the value of n which makes $(2 + 5\omega + 2\omega^2)^n = (2 + 2\omega + 5\omega^2)^n$

« $n \in \{\ldots, -6, -3, 0, 3, 6, \ldots.\}$ »

27 **Find :** $\sum_{r=0}^{10} (1 + \omega^r + \omega^{2r})$ « 12 »

28 If $z_1 = b + c\,\omega^2$, $z_2 = b\,\omega + c$, $z_3 = b\,\omega^2 + c\,\omega$

Prove that : (1) $z_1 + z_2 + z_3 = 0$ **(2)** $z_1 z_2 + z_2 z_3 + z_3 z_1 = 0$

29 If $z_1 = x + \omega y$, $z_2 = x - \omega y$

Find the value of : $z_1^3 - 3 z_1^2 z_2 + 3 z_2^2 z_1 - z_2^3$ « $8 y^3$ »

Higher skills

30 **Choose the correct answer from those given :**

(1) If $\overline{x + 2\omega + 2\omega^2} = x + 2\omega + 2\omega^2$, then $x =$ .....

(a) 1 or 2 (b) 3 or 4 (c) 0 or 3 (d) 1 or 4

(2) If $z = \frac{\sqrt{3} + i}{2}$, then $z^{69} =$ .....

(a) $-i$ (b) i (c) 1 (d) -1

(3) If $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \cdots + a_{2n} x^{2n}$

, then $a_0 + a_1 \omega + a_2 \omega^2 + \cdots + a_{2n} \omega^{2n} =$ .....

(a) zero (b) ω (c) ω^2 (d) 1

(4) If $x^2 + x + 1 = 0$

, then $\left(x + \frac{1}{x}\right)^2 + \left(x^2 + \frac{1}{x^2}\right)^2 + \left(x^3 + \frac{1}{x^3}\right)^2 + \cdots + \left(x^6 + \frac{1}{x^6}\right)^2 =$ .....

(a) 6 (b) 12 (c) 18 (d) 54

(5) $\sum_{n=1}^{12} \left(\cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3}\right) =$ .....

(a) zero (b) 1 (c) 12 (d) 78

(6) $\sum_{r=0}^{100} 100 C_r (2 + \omega^2)^{100-r} \omega^r =$ .....

(a) -1 (b) zero (c) 1 (d) 2

(7) If $z \in \mathbb{C}$, $|\omega z^2| - 5|z| = 14$, then $\left|\frac{2\bar{z} + 49}{z + 2}\right| =$ .....

(a) 7 (b) 14 (c) 21 (d) 28

31 Find the solution set of the equation : $x = 1 + \omega^x + \omega^{2x}$ « $\{3\}$ »

32 Write in the algebraic form and the trigonometric form the number z where

$z = (2\omega^2 + i)^{10} - 10 (3 i^2 - 2\omega)(2\omega^2 + i)^9$

$+ \frac{10 \times 9}{2 \times 1} (3 i^2 - 2\omega)^2 (2\omega^2 + i)^8 - \cdots + (3 i^2 - 2\omega)^{10}$ « $32 i$, $32 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$ »